

■ 2.6.11 Producing Output

The most common way to use *Mathematica* is to type in an expression, then let *Mathematica* evaluate the expression, and print the final result. Sometimes, however, you may want to get *Mathematica* to print out expressions at intermediate stages in its operation.

<code>Print[<i>expr</i>₁, <i>expr</i>₂, ...]</code>	print the <i>expr</i> _{<i>j</i>} , with no spaces in between, but with a newline (line feed) at the end
---	--

Printing expressions.

`Print` prints its arguments, with no spaces in between, but with a newline (line feed) at the end.

```
In[1]:= Print[a, b]; Print[c]
ab
c
```

This prints a table of the first five integers and their squares.

```
In[2]:= Do[Print[i, " ", i^2], {i, 5}]
1      1
2      4
3      9
4     16
5     25
```

`Print` simply takes the arguments you give, and prints them out one after the other, with no spaces in between. In many cases, you will need to print output in a more complicated format. You can do this by giving an output form as an argument to `Print`.

This prints the matrix in the form of a table.

```
In[3]:= Print[TableForm[{{1, 2}, {3, 4}}]]
1  2
3  4
```

<code>TextForm[expr]</code>	output <i>expr</i> in a form suitable for inclusion inside ordinary text
<code>StringForm["cccc``cccc", x<sub>1</sub>, x<sub>2</sub>, ...]</code>	output a string in which successive `` are replaced by successive x _i
<code>StringForm["cccc`i`cccc", x₁, ...]</code>	output a string in which each `i` is replaced by the corresponding x _i

Some output forms for generating formatted output.

`StringForm` allows you to format *Mathematica* output in a way that is somewhat analogous to the format directives used in languages like C and FORTRAN.

This generates output according to the “format string” you have given.

```
In[4]:= StringForm["x``, y``, 3, (1 + u)^2"]
Out[4]= x=3, y=(1 + u)2
```